“Volatility Modeling with GJR-GARCH, Yang–Zhang Estimation, and Macro Signals”

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Dataset & Sources

This analysis uses three categories of financial time-series data spanning January 2005 through April 2026. Daily OHLCV (Open, High, Low, Close, Volume) prices for the SPDR S&P; 500 ETF (SPY) were sourced from Yahoo Finance, providing 5,358 trading days of equity market data. The CBOE Volatility Index (VIX) was downloaded from the same source to serve as a benchmark for implied volatility. Five macroeconomic indicator series were obtained from the Federal Reserve Economic Data (FRED) database: the effective federal funds rate (DFF), the 10-year minus 2-year Treasury yield spread (T10Y2Y), the ICE BofA high yield credit spread (BAMLH0A0HYM2), the civilian unemployment rate (UNRATE), and the Consumer Price Index for all urban consumers (CPIAUCSL). All data was inspected for missing values, zero or negative prices, and OHLC consistency prior to analysis.

Data Cleaning & Preprocessing

Before any analysis, all datasets underwent a structured quality assurance process. SPY and VIX data were checked for missing OHLC values, zero or negative prices, and internal consistency violations (cases where the reported High was less than the Low). No anomalies were found across the full 5,358-day sample. The two series were confirmed to align perfectly on common trading days, requiring no date imputation or reindexing.

The five FRED macro series required more careful handling due to differences in publication frequency and coverage. The federal funds rate is reported daily, the yield spread and credit spread are reported on business days, and unemployment and CPI are monthly. To merge these onto the daily trading calendar, each series was forward-filled to propagate the most recent observation to subsequent days. This is standard practice for macro data in quantitative finance, as it avoids look-ahead bias while ensuring a complete daily panel. Missing values at the boundaries of each series were noted but did not affect the core analysis window.

Methodology

Realized volatility was estimated using the Yang–Zhang (2000) estimator, which combines overnight (close-to-open), open-to-close, and Rogers–Satchell intraday volatility components with an optimal weighting scheme that minimizes estimation variance. This approach is more statistically efficient than the standard close-to-close estimator because it incorporates the full OHLC price information available for each trading day. Rolling estimates were computed at 21-day (monthly) and 63-day (quarterly) horizons.

Conditional volatility was modeled using the GJR-GARCH(1,1,1) specification of Glosten, Jagannathan, and Runkle (1993), which extends the standard GARCH framework of Engle (1982) by adding an asymmetry parameter (γ) that captures the “leverage effect”—the empirical observation that negative return shocks tend to increase future volatility more than positive shocks of equal magnitude. Student-t distributed innovations were used to account for the heavy tails observed in the return distribution (kurtosis = 14.56, Jarque–Bera $p < 0.001$). Model selection was validated by comparing AIC and BIC across standard GARCH(1,1), GJR-GARCH(1,1,1), and EGARCH(1,1,1) specifications.

Exploratory analysis included computing rolling volatility at multiple horizons (5, 21, and 63 days), overlaying simple and exponential moving averages on the price series to identify trend regimes, and examining the autocorrelation structure of raw returns, squared returns, and absolute returns. The autocorrelation analysis confirmed that while raw returns show negligible serial dependence (consistent with weak-form market efficiency), squared and absolute returns exhibit strong, slowly decaying autocorrelation—the hallmark of volatility clustering that GARCH models are designed to capture.

To connect volatility dynamics with the broader macroeconomic environment, FRED indicators were forward-filled to daily frequency and merged with the volatility series. Correlation analysis and percentile-based regime classification (low/medium/high volatility) were used to characterize how macro conditions differ across volatility states.

Key Findings

The leverage effect is extreme and entirely one-sided. The GJR-GARCH model estimates an ARCH coefficient (α) of effectively zero and an asymmetry coefficient (γ) of 0.229, both highly significant. This means positive return shocks have no measurable impact on future volatility, while negative shocks drive volatility substantially higher. This is a stronger asymmetry than most equity studies report and suggests that for SPY, all short-term volatility dynamics are driven by downside moves. Volatility persistence is high at 0.979, indicating that shocks decay slowly.

VIX consistently overestimates realized volatility. The implied volatility index exceeds Yang–Zhang realized volatility 84.6% of the time, with a mean premium of 3.12 percentage points. This persistent volatility risk premium represents the compensation investors pay for downside protection and is consistent with well-documented findings in the options literature. The premium inverts sharply during crises (2008, 2020), when realized volatility spikes faster than implied volatility can adjust.

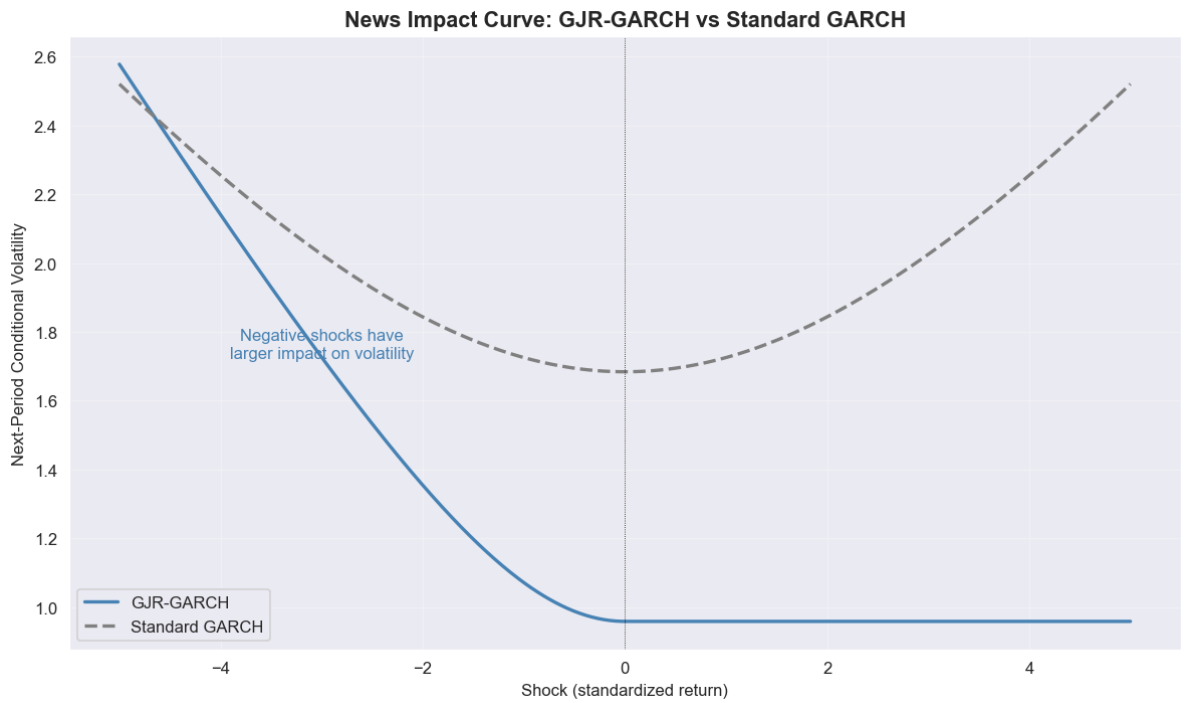

Figure 1. News impact curve comparing GJR-GARCH and standard GARCH. The GJR curve is flat for positive shocks and steep for negative shocks, illustrating the one-sided leverage effect.

Macro conditions vary meaningfully across volatility regimes. The correlation analysis reveals that the federal funds rate is negatively correlated with volatility (ρ = –0.31 to –0.68), reflecting the Fed's tendency to cut rates during periods of market stress. The 10Y–2Y yield spread is positively correlated with VIX (ρ = 0.73), as the curve steepens when the Fed eases policy. Unemployment shows a modest positive correlation with conditional volatility (ρ = 0.25), confirming that labor market deterioration coincides with elevated market uncertainty. Regime analysis further shows that high-volatility periods coincide with lower fed funds rates (4.83% vs 5.13% in low-vol regimes), a positive yield spread, and higher unemployment, consistent with the macroeconomic signature of crisis periods.

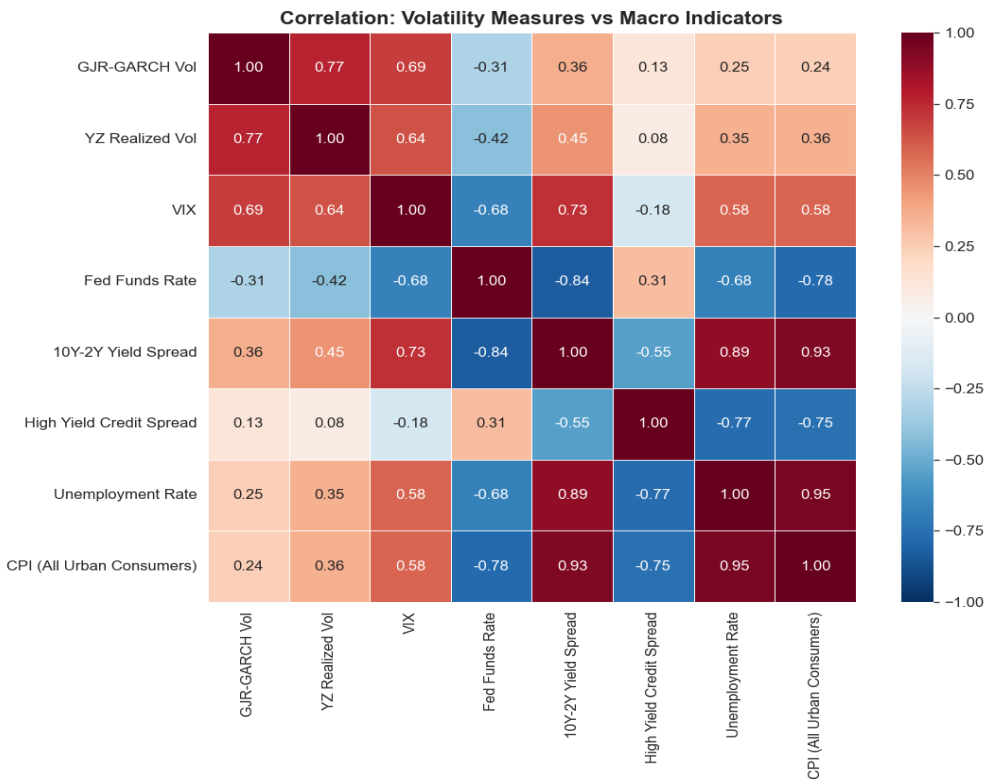

Figure 2. Correlation matrix of volatility measures (GJR-GARCH conditional, Yang–Zhang realized, VIX) and macroeconomic indicators from FRED.

Out-of-sample forecasting shows meaningful predictive power. Using a 75/25 train-test split, the GJR-GARCH model achieves a forecast-to-realized correlation of 0.733 and an RMSE of 5.09%. The Ljung–Box test on squared standardized residuals confirms that the model has adequately captured volatility clustering (p-values of 0.52, 0.72, and 0.88 at lags 10, 20, and 30 respectively).

Reflection

This project deepened my understanding of how volatility behaves in equity markets and reinforced the importance of choosing model specifications that match the empirical features of the data. The finding that the leverage effect is entirely one-sided for SPY was unexpected and pushed me to think more carefully about what drives short-term volatility dynamics. If I were to extend this work, I would explore regime-switching models (such as Markov-switching GARCH) to capture structural breaks in volatility behavior, incorporate intraday data for higher-frequency realized volatility estimation, and test whether the volatility risk premium has predictive power for future returns.

References

Engle, R. F. (1982). “Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation.” *Econometrica*, 50(4), 987–1007.

Glosten, L. R., Jagannathan, R., & Runkle, D. E. (1993). “On the Relation between the Expected Value and the Volatility of the Nominal Excess Return on Stocks.” *The Journal of Finance*, 48(5), 1779–1801.

Yang, D. & Zhang, Q. (2000). “Drift-Independent Volatility Estimation Based on High, Low, Open, and Close Prices.” *The Journal of Business*, 73(3), 477–491.